EFY TESTED
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Design Your Own Low-Cost IoT Robot

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There are many types of robots, from simple toy cars to advanced industrial robots. The Wi-Fi-controlled robot described here uses NodeMCU and Blynk app. It can be controlled wirelessly using any Wi-Fi enabled Android smartphone. The prototype wired on the breadboard is shown in Fig. 1.

Circuit and working

The block diagram of the low-cost Internet of Things (IoT) robot is shown in Fig. 2, and its circuit diagram is shown in Fig. 3. It is built around NodeMCU (board1), 5V regulator 7805 (IC1), motor driver (IC2), two DC motors (M1 and M2), 12V battery and a few other components.

The project programming of NodeMCU and interfacing Blynk app on Android device.

NodeMCU. NodeMCU is the heart of this project. Pin details of NodeMCU module are shown in Fig. 4. It is a small board compatible with breadboards and has an ESP8266 Wi-Fi controller chip operating at 2.4GHz frequency. It has micro USB port for power, programming and debugging. It also has reset and flash buttons. It can be powered using 5V via micro USB port.

Blynk app. Blynk app is used for controlling the robot from anywhere in the world. It is designed for IoT applications for smart controllers using Arduino, NodeMCU and Raspberry Pi. Various sensors including temperature, humidity, pressure and accelerometer are supported by the app.

Blynk app is simple to use, flexible and has input/output features like gauge, slide bar, joystick, etc. It has less programming but requires specific Blynk header files for interfacing. Header files and libraries are easily available on GitHub (*https://github.com/blynkkk/blynk-library*).

By adding the library in Arduino IDE, you can compile the code and program NodeMCU easily.

The joystick in Blynk app is set between 0 and 255, which are the digital reference values. When the joystick moves up and down, digital values vary between 0 and 255. When it moves along horizontal axis, voltage at x pin changes. Similarly, when it moves along vertical axis, voltage at y pin changes.

Hence, there are four directions (-x, +x, -y and +y) of joystick outputs. When the joystick moves, voltage on each output pin goes high or low depending on direction. In neutral (central) position, the joystick shows both x and y values as 128 on Blynk app.

Blynk app development

The steps for running the setup are given below.

1. Download Blynk app from Play Store and install it on your smartphone.
2. Either login or signup. You can use your current Gmail account for this.
3. Give a name to your project, say, Low-Cost IoT Robot. Select NodeMCU board from Choose Device, and Wi-Fi in Connection Type to create a new project (Fig. 5).
4. An authentication token will be sent to your registered e-mail address. Note that down as you will need it in Arduino source code (Blynkrobo.ino) later.
5. Open Arduino IDE and add the required library from *https://github.*

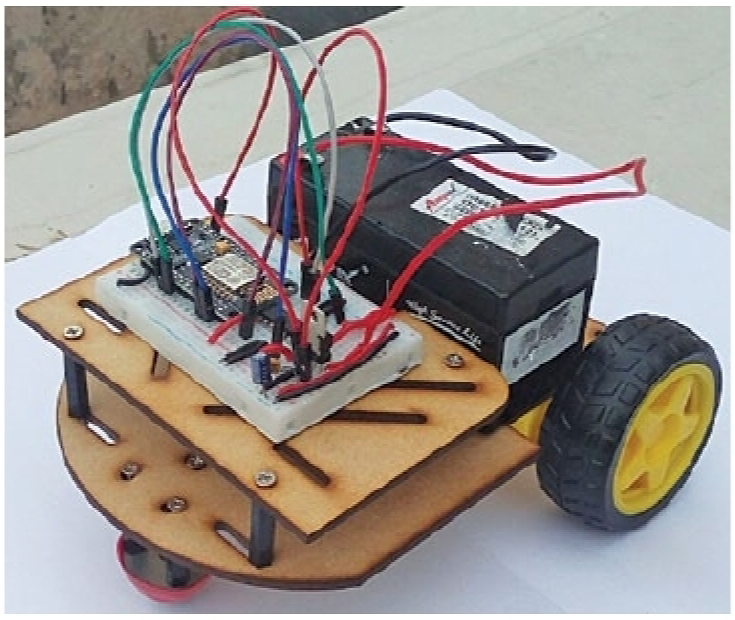

Fig. 1: IoT robot's prototype built at EFY Lab

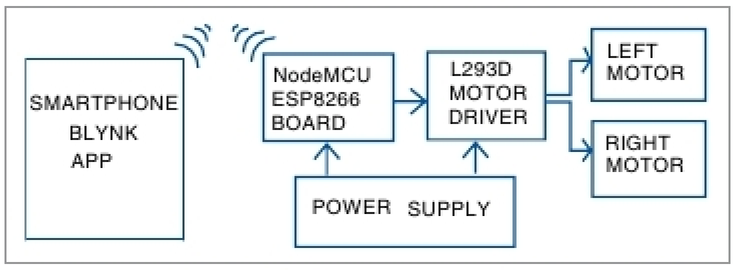

Fig. 2: Block diagram of the low-cost IoT robot

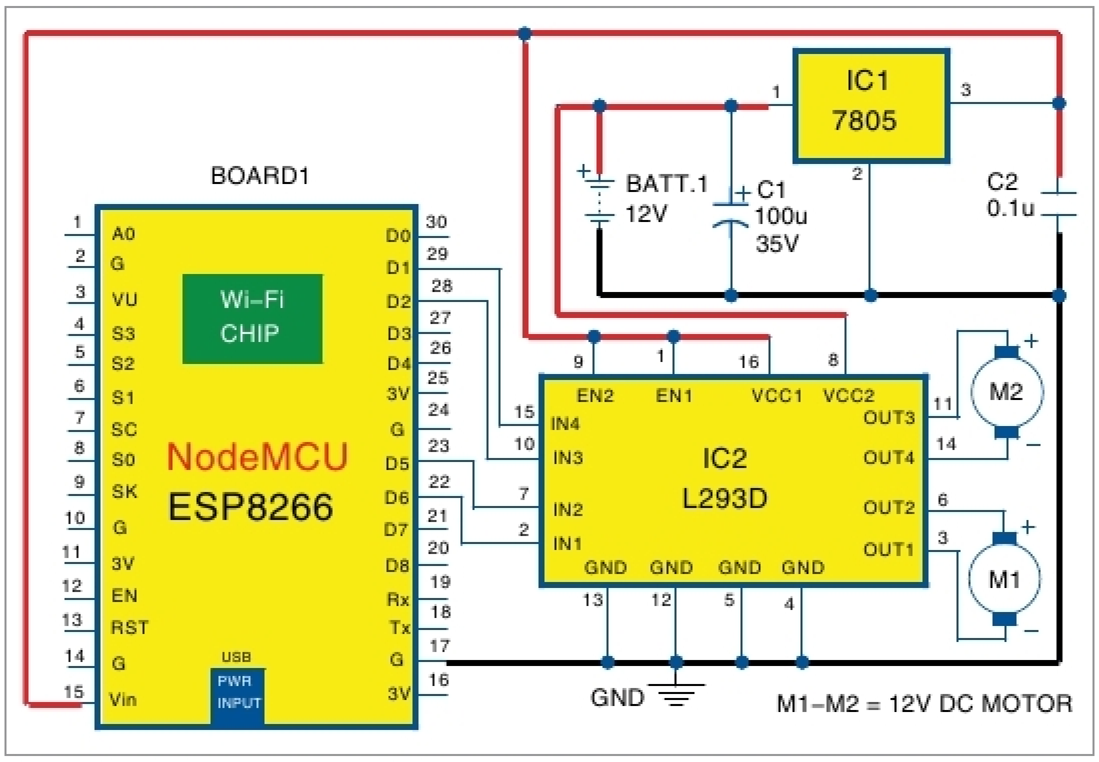

Fig. 3: Circuit diagram of the IoT robot